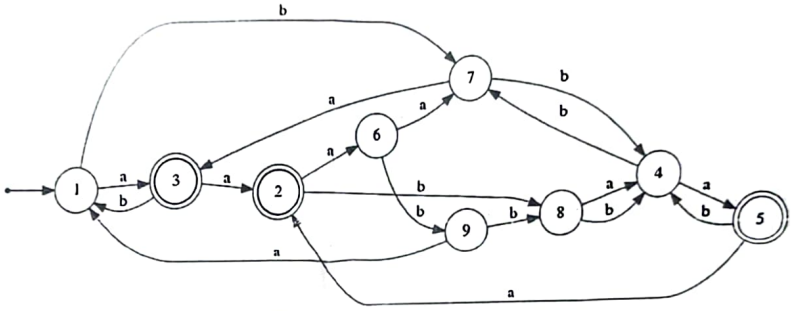
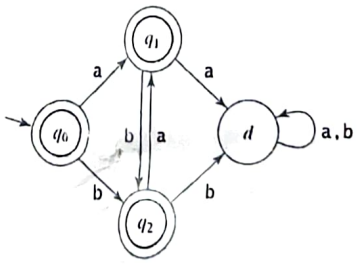
B.Tech. Comp. Engg. Vth Sem Examination 2024-25
AUTOMATA THEORY
Paper no CEN-502

Max Time : Three hours | Max. Marks: 60

Attempt any TWO parts from each question.

Write your Roll No here ..... on receipt of the question paper immediately

Q. No	CO / BL	Questions	Marks
Q1 a)	1 / L2	Explain the structure of a Finite Automaton using a suitable block diagram. Also describe its mathematical definition.	6
Q1 b)	1 / L6	Design a relevant DFA for the following expressions. i) ((a+b)*) a)* ii) ((a+aa)* + a)* Also validate your answers using suitable examples:	6
Q1 c)	1 / L3	Apply the matrix method of minimization to Minimize the following DFA.  **Verify your answers using suitable sentences.**	6
Q2 a)	2 / L6	Construct a Regular Expression for the following languages over {0,1} : i) The set of all strings containing the not containing the string '10'. ii) The set of all strings containing even no of 1's. iii) The set of all strings containing 0's in multiples of 3.	6
Q2 b)	2 / L1, L3	State and Prove the Arden's Theorem. Also apply this theorem to construct the Regular Expression for the given Automaton showing all intermediate steps.  **Verify your answers using suitable sentences.**	6

Q2 c)	2 / L1, L3	State and Define the Pumping Lemma. Also apply this lemma to analyse the Regularity of the following language : $L1 = \{a^n b^m : n>m\}$	6
Q3 a)	2 / L2, L4	Describe the relation between Language, Grammar and Automata using suitable examples? Also analyse whether the following grammar ambiguous or not? $E \rightarrow DE \mid \varepsilon$ $D \rightarrow D1 \mid 0D1 \mid 01$ **Give suitable reasons/examples to support your answer.**	6
Q3 b)	3 / L6	Construct a CFG for a language described as follows : $L_1 = \{ a^n b^m$ **, where** $m>=n>=0\}$ **Verify your answers using suitable sentences.**	6
Q3 c)	3 / L3	Convert following CFG to CNF $X \rightarrow RY \mid xRYz$ $Y \rightarrow yb \mid xZZb$ $R \rightarrow rZ \mid ZZ$ $Z \rightarrow z \mid \varepsilon$ **Mention all the steps and the corresponding rules used at appropriate positions**	6
Q4 a)	4 / L1, L2	Mathematically define a Push Down Automaton. Explain the concept of transitions between configurations of a machine.	6
Q4 b)	4 / L6	Construct a PDA with final state acceptance for the language **L** defined as: $\mathbf{L = \{ bin(n)\$mir(bin(n+1)) \mid n > 0 \} \subseteq \{0; 1; \$\}^*}$ Here *mir(n)* $\in \{0; 1\}^*$ is the mirror function and *bin(n)* $\in \{0; 1\}^*$ the binary representation (without leading zero's) of the number *'n'*. Eg. bin(10)$mir(bin(11)), where bin(10) = 1010 and mir(bin(11)) = 1011.	6
Q4 c)	4 / L3, L4	Convert the following CFG to PDA $X \rightarrow RY \mid xRYz$ $Y \rightarrow yb \mid xZZb$ $R \rightarrow rZ \mid ZZ$ $Z \rightarrow z \mid \varepsilon$ Also using the tabular approach, verify whether the string **xrybx** is acceptable by this PDA or not.	6
Q5 a)	5 / L6	Construct a Turing Machine for the followings operations: **String Reverse Append** : Given a string on the tape , the TM appends the reverse of the original string on the tape immidiately to the right of the original string; ***Verify your machine using suitable example.***	6
Q5 b)	5 / L2, L3	Explain the concept of Primitive Recursive functions. Also, prove that the function $\mathbf{f(x, y) = 2x + y}$ is primitive recursive	6
Q5 c)	5 / L2	Explain following terms in details with suitable examples: i) Halting Problem of Turing Machine ii) NP Hard Problems iii) Enumerable Languages	6

**********End of Paper**********